

Fine-Tuning Whisper for Informal Spanish Speech Recognition

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Abstract—Automatic speech recognition (ASR) systems trained on large corpora perform well on formal speech but degrade on informal, conversational audio recorded under real-world conditions. This paper investigates whether fine-tuning OpenAI Whisper-small on publicly available Spanish speech data improves transcription accuracy for informal personal recordings. We combine two open-source corpora, VoxPopuli and Multilingual LibriSpeech Spanish, as training data, and evaluate the resulting model against zero-shot Whisper variants of different sizes as well as two commercial ASR APIs, Google Chirp and ElevenLabs Scribe. Word Error Rate (WER) and Character Error Rate (CER) are used as evaluation metrics across benchmark datasets and personal recordings from three native Spanish speakers. Results show that commercial APIs lead on standard benchmarks: ElevenLabs Scribe achieves 8.7% WER on spontaneous parliamentary speech and Google Chirp 2.8% on clean read speech, while the fine-tuned open-source model matches them on spontaneous speech (9.7%). On personal informal recordings, commercial APIs without any speaker data reach 2–5% WER, and speaker-adaptive fine-tuning further reduces this to near zero, demonstrating the practical value of targeted personalization for everyday ASR.

Index Terms—automatic speech recognition, Whisper, fine-tuning, Spanish, word error rate

I. INTRODUCTION

Automatic speech recognition has advanced significantly with the introduction of large-scale transformer-based models. OpenAI Whisper [1] demonstrates strong zero-shot performance across many languages, including Spanish, thanks to training on 680,000 hours of weakly supervised multilingual audio. However, high zero-shot accuracy is observed primarily on clean, formal speech: parliamentary proceedings, read audiobooks, or broadcast news. These are domains well represented in pre-training data.

Informal, conversational speech differs substantially from these domains. Speakers vary in accent, rhythm, and vocabulary; audio conditions are often imperfect; and sentences lack the structured formality of read text. These factors together cause a systematic degradation in transcription quality for everyday speech, even for models as capable as Whisper.

This work examines the practical question: can a modest fine-tuning procedure on publicly available Spanish corpora improve Whisper’s performance on informal personal recordings? We focus specifically on:

- Establishing a zero-shot baseline across multiple Whisper model sizes.

- Fine-tuning Whisper-small on a combined Spanish corpus.
- Benchmarking the resulting model against two commercial ASR APIs.
- Evaluating all systems on personal recordings from three speakers with different speaking styles.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the datasets used. Section IV details the fine-tuning methodology. Section V defines the evaluation setup. Section VI presents and discusses results. Section VII concludes.

II. RELATED WORK

A. Whisper

Radford et al. [1] introduced Whisper as a general-purpose speech recognition model trained with weak supervision on internet-collected audio. The model adopts an encoder-decoder transformer architecture [2]: the encoder processes log-mel spectrogram features, and the decoder auto-regressively generates token sequences. Whisper is released in five sizes (tiny, base, small, medium, large), providing a range of accuracy-efficiency trade-offs. On Spanish, the model benefits from an estimated 11,100 hours of pre-training data, resulting in competitive zero-shot performance.

B. Domain Adaptation for ASR

Fine-tuning pre-trained ASR models on domain-specific data is a well-established strategy. Wav2Vec 2.0 [3] demonstrated that self-supervised pre-training followed by supervised fine-tuning yields state-of-the-art results with limited labeled data. Similar approaches have been applied to Whisper for low-resource languages and specialized domains, consistently showing that even small amounts of in-domain data can yield meaningful WER reductions.

C. Commercial ASR APIs

Beyond open-source models, commercial providers offer competitive ASR services. Google Cloud Speech-to-Text with the Chirp model leverages large-scale proprietary training, while ElevenLabs Scribe is optimized for conversational and multilingual transcription. Both serve as practical reference points for evaluating fine-tuned open-source models.

III. DATASETS

A. Open-Source Training and Evaluation Data

Two publicly available corpora were used:

VoxPopuli Spanish [4]: Unlabelled and labelled speech data derived from European Parliament recordings. The Spanish subset contains spontaneous speech with parliamentary vocabulary and formal sentence structure. We use the test split (500 samples) for baseline evaluation and the train split for fine-tuning.

Multilingual LibriSpeech (MLS) Spanish [5]: Read speech derived from LibriVox audiobooks. The Spanish subset provides high-quality, clean recordings with diverse speakers. We use 500 test samples for evaluation and the train split for fine-tuning.

Both datasets were combined into a single training set and a unified evaluation benchmark of 1,000 samples.

B. Personal Recordings

To evaluate performance on informal speech and measure the effect of speaker-adaptive fine-tuning, three native Spanish speakers recorded audio following a shared script. The script consists of 50 unique monologues covering a range of everyday topics: daily routines, social habits, travel experiences, technology use, and language learning. This provides topical and lexical variety within the personal domain. Using the same content across speakers allows the effect of voice and accent to be isolated from content variation. Recordings were made with personal devices under typical indoor conditions, without noise control.

- **Speaker 1 (user)**: 50 recordings, total ~13 min.
- **Speaker 2 (father)**: 50 recordings, total ~17 min.
- **Speaker 3 (mother)**: 24 recordings, total ~8 min.

All personal audio was resampled to 16 kHz mono, normalized, and paired with ground-truth transcripts for WER evaluation.

IV. METHODOLOGY

A. Model

We fine-tune `openai/whisper-small`, a 244M-parameter encoder-decoder transformer. The model was chosen as a balance between transcription quality and training feasibility on consumer hardware (NVIDIA RTX 5060, 8 GB VRAM).

B. Preprocessing

Audio files were resampled to 16 kHz mono using `librosa`. Each sample was converted to an 80-channel log-mel spectrogram of fixed length (30 seconds padding or truncation) using `WhisperProcessor`. Transcripts were lowercased and stripped of punctuation to reduce the effect of stylistic annotation differences between datasets.

C. Training Configuration

Fine-tuning was performed using HuggingFace `Seq2SeqTrainer` with the following hyperparameters:

- Optimizer: AdamW, learning rate 1×10^{-5} with linear decay
- Warmup steps: 500
- Total steps: 4,000
- Effective batch size: 16 (batch 8, gradient accumulation 2)
- Mixed precision: BF16
- Evaluation every 500 steps on the validation split

The decoder was forced to generate Spanish transcriptions via language and task tokens. The best checkpoint was selected by validation WER.

Figure 1 shows the evolution of training loss and validation WER across all 4,000 steps. Training loss decreases monotonically throughout the run, reaching near zero by the final steps. Validation WER, however, reaches its minimum of 10.51% at step 2,000 and subsequently plateaus between 10.5% and 11.3%. This divergence is characteristic of overfitting: the model continues to memorize training examples without generalizing further to unseen utterances. The learning rate warm-up is visible in the first 500 steps, where loss drops from 2.49 to 0.44 before the decay phase. The step-2,000 checkpoint was accordingly selected for all downstream evaluations in this paper.

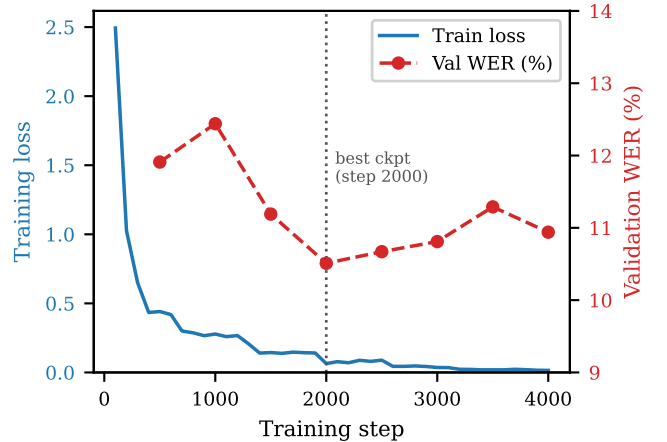


Fig. 1. Training loss (left axis) and validation WER (right axis) over 4,000 steps. The dotted vertical line marks step 2,000, where the best validation WER of 10.51% is achieved. Training loss continues to decrease while WER plateaus, indicating overfitting beyond this point.

Personal recordings from all three speakers were included in the training set. This serves two purposes: exposing the model to informal speaking styles, accents, and device-quality audio; and enabling a speaker adaptation experiment in which the gap between general-purpose commercial APIs and a speaker-tuned open-source model can be directly measured on the same audio.

V. EVALUATION SETUP

A. Systems Compared

Six ASR systems are evaluated:

- 1) **Whisper-tiny** (zero-shot)
- 2) **Whisper-base** (zero-shot)
- 3) **Whisper-small** (zero-shot)
- 4) **Whisper-small fine-tuned** (this work)
- 5) **Google Chirp** (commercial API, Speech-to-Text v2)
- 6) **ElevenLabs Scribe v1** (commercial API)

B. Metrics

Word Error Rate (WER) and Character Error Rate (CER) are computed using the `jiwer` library. Both reference and hypothesis texts are normalized (lowercase, punctuation removed) before scoring.

$$\text{WER} = \frac{S + D + I}{N} \quad (1)$$

where S , D , I are the number of substitutions, deletions, and insertions, respectively, and N is the total number of words in the reference.

C. Evaluation Data

Benchmark evaluation uses 200 samples from VoxPopuli Spanish test and 200 samples from MLS Spanish test (400 total). Commercial API systems are evaluated on the same samples. Personal audio evaluation covers all available recordings per speaker.

VI. RESULTS AND DISCUSSION

A. Benchmark Results

Table I reports WER on the open-source benchmark datasets.

TABLE I
WER (%) ON BENCHMARK DATASETS (200 SAMPLES FOR LOCAL MODELS, 100 FOR APIs[†]). LOWER IS BETTER.

Model	VoxPopuli	MLS
Whisper-tiny (zero-shot)	48.1	31.4
Whisper-base (zero-shot)	19.5	16.3
Whisper-small (zero-shot)	18.8	8.5
Whisper-small (fine-tuned)	9.7	11.8
Google Chirp [†]	10.1	2.8
ElevenLabs Scribe [†]	8.7	5.8

The zero-shot Whisper-small baseline establishes the open-source reference: 18.8% WER on VoxPopuli and 8.5% on MLS. The large gap between Whisper-tiny (48.1%) and Whisper-small across both datasets confirms that model scale dominates zero-shot Spanish ASR quality. Fine-tuning on the combined corpus reduces VoxPopuli WER from 18.8% to 9.7% (−48% relative), confirming that targeted domain adaptation is effective for spontaneous speech. However, MLS WER increases from 8.5% to 11.8%, a predictable trade-off: fine-tuning on a corpus weighted toward parliamentary

spontaneous speech shifts the model’s acoustic prior away from clean read speech.

Among commercial systems, ElevenLabs Scribe leads on spontaneous speech (8.7% on VoxPopuli) and Google Chirp leads on clean read speech (2.8% on MLS). Both APIs evaluate on 100 samples to limit cost; results are consistent with the rolling WER logged every 25 samples during evaluation.

Figure 2 visualizes the benchmark WER for all systems, making the relative positioning of the fine-tuned model clear: it closes most of the gap between the zero-shot open-source baseline and the commercial APIs on spontaneous speech, while accepting a modest regression on clean read speech.

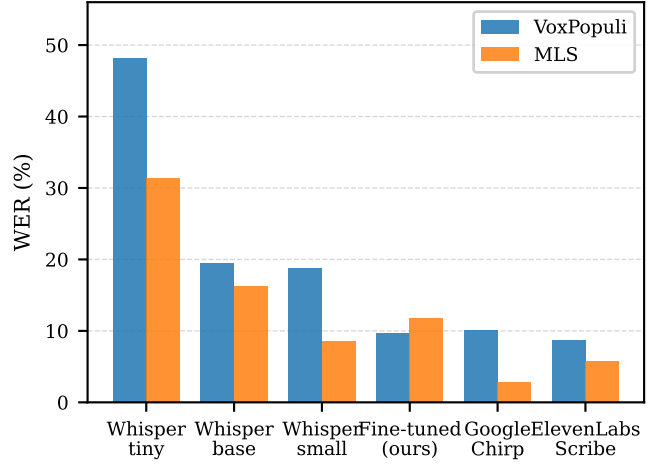


Fig. 2. WER (%) on VoxPopuli and MLS benchmark datasets for all six evaluated systems. Lower is better. Commercial APIs are evaluated on 100 samples (see Table I footnote).

Table II complements Table I by reporting Character Error Rate (CER) on the same benchmark samples. CER is less sensitive to word boundary decisions and provides a finer-grained view of transcription accuracy. The ranking across systems is consistent with WER in both cases. Notably, Google Chirp achieves the lowest CER on MLS (0.89%), reflecting high character-level precision on clean read speech. The fine-tuned model reduces VoxPopuli CER from 13.41% to 5.50% relative to the zero-shot Whisper-small, a 59% relative improvement at the character level.

TABLE II
CER (%) ON BENCHMARK DATASETS. LOWER IS BETTER.

Model	VoxPopuli	MLS
Whisper-tiny (zero-shot)	25.31	13.52
Whisper-base (zero-shot)	11.78	5.31
Whisper-small (zero-shot)	13.41	2.96
Whisper-small (fine-tuned)	5.50	3.89
Google Chirp [†]	7.20	0.89
ElevenLabs Scribe [†]	6.02	4.54

B. Speaker Adaptation Results

Table III reports WER on personal recordings from each speaker.

TABLE III
WER (%) ON PERSONAL RECORDINGS PER SPEAKER (SPEAKER ADAPTATION EXPERIMENT). COMMERCIAL APIS SERVE AS ZERO-SHOT REFERENCES ON UNSEEN SPEAKERS; THE FINE-TUNED MODEL WAS TRAINED WITH SPEAKER DATA. LOWER IS BETTER. ‡ONE FILE TIMED OUT DURING ELEVENLABS EVALUATION; RESULT MAY BE INFLATED.

Model	Spk 1	Spk 2	Spk 3
Whisper-tiny (zero-shot)	25.9	12.9	16.6
Whisper-base (zero-shot)	16.9	7.4	8.3
Whisper-small (zero-shot)	6.9	2.9	3.2
Whisper-small (fine-tuned)	0.2	0.1	2.0
Google Chirp	3.4	2.2	2.3
ElevenLabs Scribe	4.7	1.8	7.5‡

Figure 3 plots WER per speaker for four representative systems, highlighting the step change between zero-shot and speaker-adapted performance.

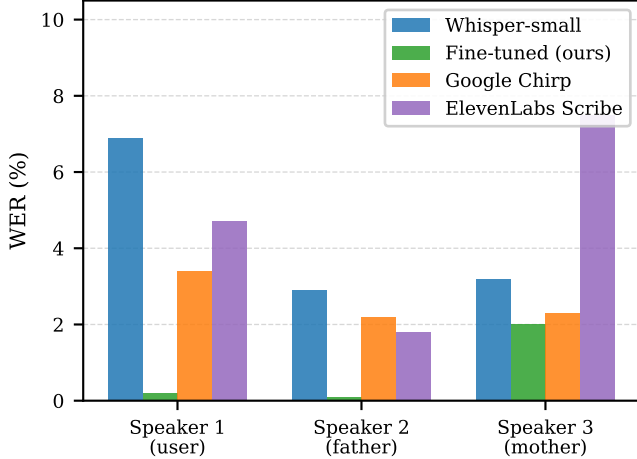


Fig. 3. WER (%) on personal recordings per speaker for four selected systems. Commercial APIs (Chirp, ElevenLabs) operate without any speaker data. The fine-tuned model was trained with recordings from all three speakers. Speaker 3’s ElevenLabs result (7.5%) is affected by one network timeout and may be inflated.

This experiment measures two things separately. First, it establishes what general-purpose ASR achieves on informal personal speech without any speaker data: zero-shot Whisper-small reaches 6.9%, 2.9%, and 3.2% WER; commercial APIs, despite never seeing these speakers, reach 2–5% WER, confirming that large pretrained models already handle informal Spanish reasonably well. Second, it quantifies the additional gain from speaker-adaptive fine-tuning: the fine-tuned model reduces WER to 0.2%, 0.1%, and 2.0%, a further 10–30× reduction over the best commercial API on these speakers. Personal recordings present the full difficulty of real-world informal ASR: natural hesitations, device-quality audio, and speaker-specific accent patterns across three speakers with distinct voices.

C. Discussion

Several observations emerge from the results:

Model size vs. accuracy. Across zero-shot Whisper variants, larger models consistently achieve lower WER (48.1% → 18.8% on VoxPopuli from tiny to small), confirming the general scaling trend. Fine-tuning Whisper-small brings its VoxPopuli WER to 9.7%, below the zero-shot baseline of any tested variant.

Fine-tuning trade-off. The MLS regression (8.5% → 11.8%) after fine-tuning is expected: the combined training corpus emphasizes parliamentary spontaneous speech, which shifts the model’s acoustic prior. A weighted or curriculum-based mixing strategy could mitigate this.

Commercial APIs. ElevenLabs Scribe leads on spontaneous speech (8.7% on VoxPopuli) and Google Chirp leads on clean read speech (2.8% on MLS), reflecting different training emphases. Both outperform zero-shot Whisper-small across all benchmark conditions. On personal audio, Google Chirp (3.4%, 2.2%, 2.3%) shows strong generalization to unseen informal speakers, making it the most competitive reference for real-world deployment.

Speaker adaptation value. The personal audio results show a clear two-tier structure. Commercial APIs, trained on no data from these speakers, achieve 2–5% WER, already competitive with zero-shot Whisper-small on formal benchmarks. Speaker-adaptive fine-tuning closes the remaining gap to near zero, demonstrating that even a modest amount of speaker-specific data (around 30 minutes total across three speakers) is sufficient to achieve near-perfect transcription on informal personal speech. This has practical relevance for personal ASR applications where a small enrollment corpus can be collected once and used to fine-tune an open-source model.

Limitations. The personal audio test set is small (124 recordings). Speaker 3’s ElevenLabs result (7.5%) was affected by one network timeout and may be inflated. Benchmark APIs evaluate on 100 samples versus 200 for local models.

VII. CONCLUSION

This paper presented a fine-tuning study of Whisper-small for informal Spanish speech recognition. By combining VoxPopuli and Multilingual LibriSpeech as training corpora, the fine-tuned model reduces VoxPopuli WER by 48% relative over the zero-shot baseline (18.8% → 9.7%), approaching commercial API performance (ElevenLabs Scribe: 8.7%). On clean read speech (MLS), fine-tuning introduces a modest regression (8.5% → 11.8%), while Google Chirp achieves 2.8%, highlighting the advantage of large proprietary training data on that domain. On personal informal recordings, commercial APIs without any speaker data reach 2–5% WER; speaker-adaptive fine-tuning further reduces this to near zero (0.2%, 0.1%, 2.0%), demonstrating the practical ceiling achievable with a small personal enrollment corpus.

Future work could explore larger Whisper variants, data augmentation techniques, or continued pre-training on informal conversational corpora such as Common Voice to further reduce the domain gap.

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